

NORA BEARTH

Analytical problem-solver with applied quantitative skills and a passion for causal machine learning, analyzing data, skiing and climbing.

EDUCATION

University of St. Gallen, Switzerland

- Ph.D. in Econometrics (defense in April 2025) 08/2021 — 04/2025
 - Dissertation: *Essays in Causal Inference*
- M.A. in Quantitative Economics and Finance (summa cum laude) 09/2019 — 09/2021
 - Exchange: *Double degree with Stockholm School of Economics*
 - Thesis: *Does childcare availability influence the gender pay gap? A natural experiment using Swiss cantons*
- B.A. in Business Administration (summa cum laude) 09/2015 — 09/2018
 - Exchange: *University of Western Australia in Perth*

Stockholm School of Economics, Sweden

- M.Sc. in Applied Economic Analysis (summa cum laude) 09/2019 — 09/2021
 - Exchange: *Double degree with University of St. Gallen*

WORK EXPERIENCE

- Doctoral Research Assistant — Swiss Institute for Empirical Economic Research, University of St. Gallen** 08/2021 — now
 - Derived, proved and implemented new method for estimating differences in treatment effects using double/debiased machine learning.
 - Cleaned and analyzed various large datasets from the government and from health insurers.
 - Taught exercise classes in data analytics, microeconomics and macroeconomics; consistently received very good course evaluations.

- Intern M&A Advisory — KMPG AG, Zurich** 03/2019 — 07/2019
 - Contributed to M&A publications (Clarity on M&A, M&A Review).
 - Worked on live M&A mandates and pitches.
 - Performed valuations, industry researches, market analyses and company research.

- Intern Corporate Tax — KPMG AG, Zurich** 09/2018 — 02/2019
 - Tax compliance: filing tax returns, checking tax invoices, overviews in excel.
 - Tax advisory: due diligence (tax for M&A), legal clarifications, assistance with tax advice regarding reorganization and market entry.

- Part time secretary — Kunz Schmid Rechtsanwälte und Notare AG, Chur** 06/2016 — 03/2017
 - Independent management of the secretariat and correspondence with clients, counterparties and authorities.
 - Assisting lawyers in drafting contracts and notarial deeds.

RELEVANT ACADEMIC RESEARCH

- Causal Machine Learning for Moderation Analysis** 📄 🌐
 - We introduce a new parameter of interest to estimate differences in treatment effects, show how to estimate it using double/debiased machine learning (DML) and proof its asymptotic properties.
 - Revise and resubmit at Journal of Business and Economic Statistics (joint with M. Lechner).*

- Improving the Finite Sample Estimation of Average Treatment Effects using Double/Debiased Machine Learning with Propensity Score Calibration** 📄 🌐
 - We show that calibrating propensity scores improves the estimation of average treatment effects when using DML.
 - Under revision at the Journal of Applied Econometrics (joint with D. Ballinari).*

- Beyond Baby Blues: The Child Penalty in Mental Health in Switzerland** 📄
 - I analyze the mental health penalty for women after childbirth in Switzerland leveraging insurance data.
 - Working paper.*

- The Heterogeneous Effects of Active Labour Market Policies in Switzerland** 📄
 - We analyze whether the Swiss active labor market programmes increase future employment and earnings of the unemployed by using causal machine learning methods and leveraging an administrative datasets.
 - Working paper (joint with F. Mascolo, F. Muny, M. Lechner and J. Mareckova).*

RELEVANT SKILLS & EXPERTISE

- Technical Skills:** Advanced: Python (NumPy, scikit-learn, Pandas, statsmodels), R, Latex; Basic: SQL, Git.
- Languages:** Raetho-Romansh, German (bilingual), English (proficiency), French (beginner), Italian (beginner)
- Talks and Paper Presentations:** Annual congress of Swiss Society of Economics and Statistics, COMPIE conference, Annual meeting Verein für Sozialpolitik, CSS, Swica, Mobiliar, University of Lucern, University of St. Gallen

EXTRACURRICULAR ACTIVITIES

- Founder member — Millenial Board Switzerland, Chur 04/2018 — 04/2023
- Participant — Econometric Games, Amsterdam 04/2021 + 04/2022
- Student representative SEPS and member of student parliament — SHSG, St. Gallen 06/2019 — 05/2020
- Part-time Ski instructor — Skiclub and Ski- und Snowboard Schule, Obersaxen 12/2016 — 04/2020
- President — Orchestra of the University, St. Gallen 06/2016 — 05/2017